



**COMPETENCY STANDARD
FOR
PRODUCTIVITY IMPROVEMENT AND QUALITY CONTROL
IN YARN
(TEXTILE SECTOR)**

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance
Government of the People's Republic of Bangladesh

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The Competency Standard for **Productivity Improvement and Quality Control in Yarn** is designed to guide the development of curricula, teaching materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry, so that individuals who pass through the set standard via assessment would be qualified and settled for a relevant job.

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades, which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The textile sector is the largest manufacturing and export-earning sector of Bangladesh, and yarn manufacturing (spinning) is the primary value-adding stage of the textile value chain. In a spinning mill, however, installed capacity alone does not determine competitiveness. Profitability and market acceptance depend on how efficiently the machines and manpower are utilized, how much of the raw material is converted into saleable yarn, and how consistently the yarn meets the quality characteristics demanded by weaving, knitting, dyeing and export buyers. A small improvement in machine efficiency, waste percentage or yarn realization, or a small reduction in yarn faults and contamination, translates directly into significant cost savings and higher value realization for the mill.

Bangladeshi spinning mills continue to report avoidable losses arising from high end breakage rates, excessive waste, unplanned downtime, contamination, count variation and customer complaints. These losses persist largely because there is a shortage of workers trained specifically to measure, analyse and improve productivity and quality using structured techniques and standard test methods. Existing training programs concentrate mainly on machine operation, while the analytical and laboratory competencies required for process control, testing, waste control, statistical process control and continuous improvement have remained largely un-standardized.

This Competency Standard has therefore been developed to prepare a competent workforce for productivity improvement and quality control in yarn manufacturing. It covers assessment of raw material quality, in-process quality control from mixing to roving, yarn testing and evaluation, identification and control of yarn faults and contamination, measurement and improvement of machine and manpower productivity, waste control and cost reduction, application of quality management systems and continuous improvement tools, and monitoring of the process environment, machine condition and instrument calibration.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and to ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competency and its corresponding Elements.

OVERVIEW:

A Competency Standard is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

1. provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
2. enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies.
3. encourages the development and delivery of flexible training that suits individual and industry requirements.
4. encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standards are developed by a working group comprised of national and international process experts and the participation of experts from the industry to identify the competencies required of an occupation in a particular sector.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at a training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

1. Reference to Industry Sector, Occupational Title and Occupational Description
2. Unit code
3. Unit title
4. Unit descriptor
5. Unit of Competency
6. Elements and performance criteria
7. Variables and range statement
8. Evidence guides

Together all the parts of a Unit of Competency:

1. Describe a work activity
2. Guide the assessor in determining whether the candidate is competent.

This Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competency:

1. An overview of all Units of Competency for the occupation and their corresponding duration required for completion of training.
2. The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

DESCRIPTION OF THE OCCUPATION AND THE TRAINING COURSE:

Industry Sector	Textile
Occupation Title	Productivity Improvement and Quality Control in Yarn
Occupational Description	A worker competent in Productivity Improvement and Quality Control in Yarn monitors, measures, analyses and improves the productivity and quality performance of a yarn manufacturing (spinning) mill. The occupation covers assessment of fibre and raw material quality; in-process quality control from mixing and blow room through carding, drawing, combing and roving; yarn quality testing and evaluation in the laboratory; identification and control of yarn faults and contamination; measurement and improvement of machine, material and manpower productivity; waste control, yarn realization and cost reduction; application of quality management system requirements and continuous improvement tools; and monitoring of the process environment, machine condition and instrument calibration. The worker performs these tasks under general supervision, following standard operating procedures, standard test methods, control plans, machine setting sheets, safety requirements and environmental practices of the mill, and records, analyses and reports productivity and quality data using workplace formats and basic digital tools.
Level of the Course	Basic Level Skills Training
Nominal Duration	360 Hours (03 Months)
Entry Qualification of Trainees	Grade X (SSC) passed or equivalent; candidates having a Diploma in Textile Engineering or prior working experience in a spinning mill will be given preference; physically and mentally fit to work in a spinning mill environment; minimum age as per the prevailing labour law of Bangladesh.
Target Number of Trainees	A total of 1,500 trainees to be trained by the end of 2028
Training Providers	Textile industries and training institutes having the capacity to deliver and run the training program in accordance with this Competency Standard and the Workshop/Lab Facility Standard.
Mode of Delivery	Institution-based and/or industry-based competency-based training combining theory, demonstration, guided practice, laboratory testing and practical performance in an actual or simulated workplace.
Employment / Job Placement	The training providers (industries and training institutes) will arrange employment for at least 65% of the total enrollees upon completion of the training.
Assessment and Certification	Competency assessment will be conducted as per the Assessment Evidence Guide of each unit, in an assessment/training centre or in an actual or simulated

	workplace, by a nationally certified assessor or occupation-specific industry expert.
Possible Job Titles	Quality Control Assistant/Inspector (Spinning), Yarn Testing Laboratory Technician, In-process Quality Checker, Process Control Assistant, Productivity/Work Study Assistant, Industrial Engineering Assistant, Waste Control Assistant, Quality Assurance Assistant, Shift Quality Supervisor (trainee level).

UNITS & ELEMENTS AT A GLANCE:

Generic Competencies (20 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TEX-PIQCY-01-G	Apply Occupational Health and Safety (OHS) Practices in the Workplace	1. Identify OHS policies and procedures 2. Apply personal health and safety practices 3. Report hazards and risks 4. Respond to emergencies	10
SICIP-TEX-PIQCY-02-G	Work in a Team Environment	1. Identify team goals and processes 2. Communicate and cooperate with team members 3. Work as a team member 4. Solve problems as a team member	10
Total Hours			20 Hrs.

Sector Specific Competencies (20 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TEX-PIQCY-01-S	Work in the Textile Sector	1. Identify the textile sector and its main activities 2. Understand basic roles in textile production processes 3. Recognize common tools and equipment used 4. Understand simple quality requirements in textile work	10
SICIP-TEX-PIQCY-02-S	Apply Green Practices in the Textile Sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
Total Hours			20 Hrs.

Occupation Specific Competencies (320 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TEX-PIQCY-01-O	Interpret Productivity and Quality Concepts in Yarn Manufacturing	1. Interpret productivity concepts and measures 2. Interpret quality concepts and standards 3. Identify factors affecting productivity and quality 4. Apply basic technical calculations used in a spinning mill	35
SICIP-TEX-PIQCY-02-O	Perform Fibre and Raw Material Quality Assessment	1. Prepare for fibre and raw material testing 2. Collect and condition samples 3. Test fibre properties 4. Interpret fibre test results 5. Record and report fibre quality data	40
SICIP-TEX-PIQCY-03-O	Perform In-process Quality Control in Spinning Preparatory Processes	1. Prepare for in-process quality control 2. Check blow room, lap and chute feed quality 3. Check card sliver quality 4. Check draw frame, comber and roving quality 5. Record and report in-process quality data	45
SICIP-TEX-PIQCY-04-O	Perform Yarn Quality Testing and Evaluation	1. Prepare the laboratory and instruments for yarn testing 2. Test yarn count, twist and moisture regain 3. Test yarn strength and elongation 4. Test yarn evenness, imperfections and hairiness 5. Interpret, record and report yarn test results	45
SICIP-TEX-PIQCY-05-O	Identify and Control Yarn Faults and Contamination	1. Identify yarn, package and process faults 2. Analyse causes of faults 3. Control contamination and foreign fibre 4. Apply corrective and preventive actions	35

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TEX-PIQCY-06-O	Measure and Improve Machine and Manpower Productivity	1. Collect production and performance data 2. Calculate production, efficiency and utilization 3. Analyse downtime, breakages and losses 4. Monitor manpower productivity and work assignment	45
SICIP-TEX-PIQCY-07-O	Apply Waste Control and Cost Reduction Techniques	1. Identify and classify waste generated in the spinning process 2. Measure and calculate waste and yarn realization 3. Analyse causes of excess waste 4. Apply waste control and reuse measures 5. Report waste and cost saving	30
SICIP-TEX-PIQCY-08-O	Apply Quality Management System	1. Interpret quality management system requirements 2. Participate in quality audit and documentation 3. Prepare and present improvement reports	25
SICIP-TEX-PIQCY-09-O	Monitor Process Environment, Machine Condition and Calibration	1. Monitor atmospheric and workplace conditions 2. Monitor machine condition and critical settings 3. Verify calibration and care of testing instruments 4. Maintain records	20
Total Hours			320 Hrs.

Grand Total Duration of the Training Course	360 Hours (03 Months)
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COMPETENCY STANDARD: PRODUCTIVITY IMPROVEMENT AND QUALITY CONTROL IN YARN

The Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE	Nominal Duration: 10 Hrs.	Unit Code: SICIP-TEX-PIQCY-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply Occupational Health and Safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures; applying personal health and safety practices; reporting hazards and risks; and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1. 1 <u>OHS policies</u> and safe operating procedures are read and understood. 1. 2 Safety signs and symbols are identified and followed. 1. 3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Apply personal health and safety practices	2. 1 OHS policies and procedures are followed and practiced. 2. 2 <u>Personal Protective Equipment (PPE)</u> is selected and used. 2. 3 Personal hygiene is maintained.
3. Report hazards and risks	3.1 <u>Hazards and risks</u> are identified, assessed and controlled. 3.2 Incidents arising from hazards and risks are reported to the authority. 3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.
4. Respond to emergencies	4. 1 Alarms and warning devices are responded to. 4. 2 <u>Emergency response plans and procedures</u> are implemented. 4. 3 <u>First aid procedure</u> is applied during emergency situations.

Range of Variables:

Variable	Range (May include but not limited to)
1. OHS policies	1. 1 International OHS requirements 1. 2 Bangladesh standards for OHS 1. 3 Bangladesh Labour Act and Rules 1. 4 Building Code 1. 5 Fire Safety Rules and Regulations 1. 6 Industry guidelines and mill safety manual
2. Personal Protective Equipment (PPE)	2. 1 Nose/face mask (dust mask) 2. 2 Ear plugs or ear muffs 2. 3 Hand gloves 2. 4 Safety shoes 2. 5 Head cap/helmet 2. 6 Apron and overalls 2. 7 Goggles and safety glasses 2. 8 Hair net
3. Hazards and risks	3.1 Physical hazards (moving machine parts, nip points, noise, vibration, heat) 3.2 Chemical hazards (lubricants, cleaning agents, solvents) 3.3 Biological hazards (bacteria, fungus in damp areas) 3.4 Ergonomic hazards (repetitive work, standing, manual lifting) 3.5 Electrical hazards 3.6 Fire hazards (cotton dust, fly accumulation, static electricity)
4. Emergency response plans and procedures	4.1 Firefighting procedures 4.2 Earthquake response procedures 4.3 Evacuation procedures 4.4 Medical and first aid 4.5 Electrical accident response 4.6 Emergency machine stop procedures
5. First aid procedure	5. 1 Washing of open wound 5. 2 Washing chemically infected area 5. 3 Applying bandage 5. 4 Tourniquet 5. 5 Applying CPR (Cardiopulmonary Resuscitation) 5. 6 Taking appropriate medicine

Curricular Content Guide:

1. Underpinning Knowledge	1.1 OHS workplace policies and procedures. 1.2 Work safety procedures in a textile mill. 1.3 Emergency procedures. 1.3.1 Firefighting 1.3.2 Earthquake response.
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	1.3.3 Explosion response. 1.3.4 Accident response. 1.4 Types of hazards (biological, chemical, physical, ergonomic and electrical) and their effects. 1.5 PPE types and uses. 1.6 Personal hygiene practices. 1.7 Safety signs, symbols and colour codes. 1.8 OHS awareness and reporting requirements.
2. Underpinning Skills	2. 1 Identifying OHS policies and procedures 2. 2 Following personal work safety practices 2. 3 Selecting and using appropriate PPE 2. 4 Reporting hazards and risks 2. 5 Responding to emergency procedures 2. 6 Maintaining physical well-being in the workplace 2. 7 Performing basic firefighting using fire extinguishers 2. 8 Applying basic first aid procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 PPEs 4.3 Firefighting equipment 4.4 Emergency response manual 4.5 Safety signs and symbols 4.6 First aid kits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Followed OHS policies and procedures. 1.2 Selected and used personal protective equipment (PPE). 1.3 Reported incidents arising from hazards and risks to the authority. 1.4 Implemented emergency response plans and procedures. 1.5 Applied basic first aid procedure.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning

	2. 4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: WORK IN A TEAM ENVIRONMENT	Nominal Duration: 10 Hrs.	Unit Code: SICIP-TEX-PIQCY-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to work in a team environment. It specifically includes the tasks of identifying team goals and processes, communicating and cooperating with team members, working as a team member and solving problems as a team member.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and processes	1.1 Team goals and collaborative decision-making processes are identified. 1.2 Roles and responsibilities of team members are identified. 1.3 Relationships within the team and with other workers are identified.
2. Communicate and cooperate with team members	2. 1 Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives. 2. 2 Formal and informal <u>forms of communication</u> are used effectively to support team achievement. 2. 3 Diversity in character is respected and valued in team functioning. 2. 4 Views and opinions of other team members are understood and valued. 2. 5 Workplace terminology is used correctly to assist communication.
3. Work as a team member	3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with the team. 3.2 Tasks are performed in accordance with organizational and team requirements, specifications, and workplace procedures. 3.3 Support to other team members is provided to ensure the team achieves its goals. 3.4 Agreed reporting lines are followed using standard operating procedures.
4. Solve problems as a team member	4.1 Current and potential problems faced by the team are identified. 4.2 A solution to the problem is identified. 4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated.

Range of Variables:

Variable	Range (May include but not limited to)
1. Forms of communication	1. 1 Agenda 1. 2 Simple reports such as progress and incident reports 1. 3 Job sheets and shift handover log 1. 4 Operational manuals and SOPs 1. 5 Brochures and promotional material 1. 6 Visual and graphic materials 1. 7 Standards 1. 8 OSH information 1. 9 Signs and symbols

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Team goals and collaborative decision-making processes 1. 2 Roles and responsibilities of team members 1. 3 Relationships within the team and with other workers 1. 4 Effective interpersonal skills to interact with team members 1. 5 Effective formal and informal forms of communication 1. 6 Value of diversity in team functioning 1. 7 Correct use of workplace terminology 1. 8 Team's duties, responsibilities, authorities, objectives and task requirements 1. 9 Support mechanisms for other members of the team to ensure achievement of goals 1. 10 Methods of identifying current and potential problems faced by a team 1. 11 Effective problem-solving methods and evaluation of outcomes
2. Underpinning Skills	2. 1 Identifying team goals and collaborative decision-making processes 2. 2 Identifying roles and responsibilities of team members 2. 3 Using effective interpersonal skills to interact with team members 2. 4 Using formal and informal forms of communication 2. 5 Understanding and valuing views and opinions of other team members 2. 6 Performing tasks in accordance with organizational and team requirements, specifications and workplace procedures 2. 7 Supporting other members of the team to ensure the team achieves its goals 2. 8 Identifying current and potential problems faced by the team 2. 9 Identifying solutions to problems

	2. 10 Solving problems effectively and evaluating the outcome of the implemented solution
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Manuals/Guidelines 4.3 Pens 4.4 Papers 4.5 Workbooks

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Identified team goals and work processes. 1. 2 Communicated and cooperated with team members. 1. 3 Worked as a team member. 1. 4 Solved problems as a team member.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 4. Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: WORK IN THE TEXTILE SECTOR	Nominal Duration: 10 Hrs.	Unit Code: SICIP-TEX-PIQCY-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for work in the textile sector. It specifically includes the tasks of identifying the textile sector and its main activities, understanding basic roles in textile production processes, recognizing common tools and equipment used, and understanding simple quality requirements in textile work.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify the textile sector and its main activities	1. 1 The textile sector is identified within the context of national and global industries. 1. 2 <u>Main activities</u> of the textile sector are recognized. 1. 3 <u>Different stages of the textile value chain</u> are described in simple terms. 1. 4 Key workplaces in the textile sector are identified. 1. 5 Importance of the textile sector in the economy and the environment is understood.
2. Understand basic roles in textile production processes	2. 1 <u>Basic job roles in textile production</u> are identified and described. 2. 2 Simple production workflows and task sequences are understood. 2. 3 Teamwork and coordination in production processes are identified. 2. 4 Workplace safety roles and responsibilities are understood and followed. 2. 5 Shift working practices and reporting lines are understood.
3. Recognize common tools and equipment used	3.1 <u>Common tools used in textile processing</u> are identified correctly. 3.2 <u>Basic machines and equipment used in production</u> are recognized. 3.3 Functions of tools and equipment are understood in simple terms. 3.4 Basic maintenance and cleaning needs of tools and equipment are recognized.
4. Understand simple quality requirements in textile work	4.1 Basic quality standards in textile work are identified. 4.2 <u>Simple defects in textile products</u> are recognized. 4.3 Importance of maintaining quality during production is understood.

Elements of Competency	Performance Criteria
	4.4 Work output is checked against basic quality expectations. 4.5 Good workmanship practices are followed to ensure quality.

Range of Variables:

Variable	Range (May include but not limited to)
1. Main activities	1. 1 Fibre production and preparation 1. 2 Yarn manufacturing (spinning) 1. 3 Fabric manufacturing (weaving and knitting) 1. 4 Wet processing (dyeing, printing and finishing) 1. 5 Garment manufacturing 1. 6 Testing and quality assurance 1. 7 Packaging, storage and export
2. Different stages of the textile value chain	2. 1 Natural and man-made fibre production 2. 2 Fibre preparation and mixing 2. 3 Yarn manufacturing (spinning) 2. 4 Fabric manufacturing (weaving/knitting) 2. 5 Dyeing, printing and finishing 2. 6 Cutting, sewing and finishing of garments 2. 7 Packaging, distribution and export
3. Basic job roles in textile production	3.1 Blow room operator/helper 3.2 Card tenter 3.3 Draw frame and comber operator 3.4 Simplex (speed frame) operator 3.5 Ring frame sider and doffer 3.6 Winding and packing operator 3.7 Quality checker/laboratory assistant 3.8 Maintenance fitter/electrician 3.9 Line leader/supervisor
4. Common tools used in textile processing	4.1 Sliver and roving cutter, scissors and nippers 4.2 Hooks, brushes and dusting tools 4.3 Spanner, screwdriver and plier set 4.4 Oil can and grease gun 4.5 Feeler and setting gauges 4.6 Wrap reel, wrap block and electronic balance 4.7 Tachometer and hand lens 4.8 Vacuum cleaner and air blowing gun
5. Basic machines and equipment used in production	5. 1 Bale opener/plucker and blow room line 5. 2 Carding machine 5. 3 Draw frame (breaker and finisher) 5. 4 Lap former and comber 5. 5 Speed frame (simplex)

Variable	Range (May include but not limited to)
	5. 6 Ring frame 5. 7 Winding machine/autoconer 5. 8 Humidification and waste collection plant 5. 9 Testing laboratory equipment
6. Simple defects in textile products	6. 1 Neps and slubs 6. 2 Thick and thin places 6. 3 Count variation and uneven yarn 6. 4 High hairiness 6. 5 Foreign fibre and contamination 6. 6 Sliver and roving breakage 6. 7 Soft/hard and damaged packages 6. 8 Oil and dirt stains

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Structure and importance of the textile sector in the national and global economy. 1. 2 Main activities and stages of the textile value chain. 1. 3 Position of yarn manufacturing within the textile value chain. 1. 4 Basic job roles, responsibilities and reporting lines in textile production. 1. 5 Simple production workflows, task sequences and shift practices. 1. 6 Common tools, machines and equipment used in textile production and their functions. 1. 7 Basic maintenance and cleaning needs of tools and equipment. 1. 8 Basic quality standards and common defects in textile products. 1. 9 Importance of quality, productivity and good workmanship in the workplace.
2. Underpinning Skills	2. 1 Identifying the textile sector and its main activities 2. 2 Describing the stages of the textile value chain in simple terms 2. 3 Understanding basic job roles and responsibilities in textile production processes 2. 4 Following simple production workflows, task sequences and teamwork practices 2. 5 Applying workplace safety roles and responsibilities in production activities 2. 6 Recognizing common tools, machines and equipment and their functions 2. 7 Recognizing basic maintenance and cleaning needs of tools and equipment

	2. 8 Recognizing simple defects and checking work output against basic quality requirements 2. 9 Following good workmanship practices to ensure consistent quality output
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Charts, models and samples of fibre, sliver, roving and yarn 4.3 Multimedia projector, laptop and audio-visual clips 4.4 Learning manual and handouts 4.5 Stationery

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Identified the textile sector and its main activities. 1. 2 Understood basic roles in textile production processes. 1. 3 Recognized common tools and equipment used. 1. 4 Understood simple quality requirements in textile work.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY GREEN PRACTICES IN THE TEXTILE SECTOR	Nominal Duration: 10 Hrs.	Unit Code: SICIP-TEX-PIQCY-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply green practices in the textile sector. It specifically includes the tasks of interpreting green concepts, minimizing resource use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1. 1 Basic concepts of environmental sustainability and green practices are explained clearly. 1. 2 <u>Environmental impacts of textile production</u> processes are identified. 1. 3 <u>Resource efficiency concepts</u> are interpreted. 1. 4 Waste reduction, reuse and recycling principles are described in relation to yarn manufacturing operations. 1. 5 <u>Eco-friendly practices in the textile sector</u> are recognized and related to workplace activities.
2. Minimize resource use in the workplace	2. 1 Electricity, water, compressed air and raw materials are used efficiently according to workplace requirements. 2. 2 <u>Unnecessary consumption of resources</u> is identified and avoided during operations. 2. 3 Machines and equipment are operated in energy-efficient modes where applicable. 2. 4 Material wastage is minimized through careful handling and accurate measurement. 2. 5 <u>Resource-saving practices</u> are applied where possible. 2. 6 Leakage, spillage and wastage of materials are promptly controlled and reported. 2. 7 <u>Workplace practices</u> for resource conservation are followed.
3. Implement waste management practices	3.1 <u>Types of waste</u> generated in yarn manufacturing operations are identified correctly. 3.2 Waste materials are segregated according to workplace and environmental requirements. 3.3 <u>Reusable and recyclable materials</u> are separated from non-recyclable waste. 3.4 Waste is stored safely in designated containers or areas. 3.5 Waste disposal is carried out following environmental rules and organizational procedures.

Elements of Competency	Performance Criteria
	3.6 Hazardous waste is handled carefully according to safety guidelines. 3.7 Waste reduction practices are applied to minimize environmental impact during operations.

Range of Variables:

Variable	Range (May include but not limited to)
1. Environmental impacts of textile production	1. 1 High electrical energy consumption of spinning machinery and humidification plants 1. 2 Cotton dust, fibre fly and micro-fibre emission 1. 3 Noise pollution from machines and compressors 1. 4 Solid waste generation (droppings, comber noil, soft and hard waste) 1. 5 Water consumption and effluent from wet processing 1. 6 Air emission and greenhouse gases from boilers and generators 1. 7 Used oil, grease and contaminated cotton waste
2. Resource efficiency concepts	2. 1 Energy-efficient motors and inverter/variable frequency drives 2. 2 Control of compressed air leakage and optimized air pressure 2. 3 Optimized humidification and ventilation control 2. 4 Reduction of soft and hard waste percentage 2. 5 Preventive maintenance to reduce power consumption 2. 6 Energy-efficient lighting and switching-off practices 2. 7 Optimum machine loading and reduction of idle running
3. Eco-friendly practices in the textile sector	3.1 Recycling of soft waste back into mixing where permitted 3.2 Use of recycled, organic or sustainably sourced fibre 3.3 Dust collection, filter and waste recovery systems 3.4 Compliance with Department of Environment (DoE) rules and regulations 3.5 Effluent treatment plant (ETP) and rainwater harvesting 3.6 Reuse of cans, bobbins, cones, cartons and packaging materials 3.7 Green building and energy audit practices
4. Unnecessary consumption of resources	4.1 Idle running of machines and motors 4.2 Compressed air leakage in pipelines and guns 4.3 Excess lighting in unoccupied areas 4.4 Over-cleaning with air guns instead of vacuum 4.5 Excess lubricant and cleaning material use 4.6 Excessive good fibre loss with droppings and noil

Variable	Range (May include but not limited to)
5. Resource-saving practices	5. 1 Switching off machines, lights and fans when not required 5. 2 Reporting air, water and oil leakage promptly 5. 3 Using vacuum cleaning instead of air blowing where possible 5. 4 Reusing cans, tubes, cones and packaging materials 5. 5 Maintaining correct machine settings to reduce waste 5. 6 Recording and monitoring waste and energy use
6. Workplace practices	6. 1 Standard operating procedures (SOPs) 6. 2 Environmental policy of the organization 6. 3 Housekeeping and 5S practices 6. 4 Energy and waste monitoring records 6. 5 Instructions from supervisors and safety notices
7. Types of waste	7.1 Blow room droppings and card droppings/flat strips 7.2 Comber noil 7.3 Sliver, roving and pneumafil soft waste 7.4 Hard waste from winding (yarn and thread waste) 7.5 Damaged cans, bobbins, cones and tubes 7.6 Packaging waste (poly bags, bale wrappers, straps, cartons) 7.7 Used oil, grease and oil-contaminated cotton (hazardous waste) 7.8 Used filters, sweeps and floor waste
8. Reusable and recyclable materials	8.1 Usable soft waste for back-mixing 8.2 Comber noil for coarse count and open-end spinning 8.3 Cans, bobbins, tubes and cones 8.4 Cartons, poly bags and bale straps 8.5 Metal and plastic scrap 8.6 Paper and stationery waste

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Basic concepts of environmental sustainability, green practices and climate-resilient production. 1. 2 Environmental impacts of textile and yarn manufacturing operations. 1. 3 Resource efficiency concepts relating to energy, water, compressed air and raw material. 1. 4 Principles of waste reduction, reuse and recycling in a spinning mill. 1. 5 Eco-friendly practices and technologies applied in the textile sector. 1. 6 Types of waste generated at each stage of yarn manufacturing and permissible waste levels.
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	1. 7 Waste segregation, storage, disposal methods and compliance with environmental regulations. 1. 8 Hazardous waste handling and workplace environmental safety procedures. 1. 9 Relevant national environmental rules and organizational environmental policy.
2. Underpinning Skills	2. 1 Interpreting green concepts and relating them to workplace activities 2. 2 Identifying environmental impacts of yarn manufacturing operations 2. 3 Applying resource efficiency principles in workplace operations 2. 4 Using electricity, compressed air, water and raw materials efficiently 2. 5 Identifying and avoiding unnecessary consumption of resources 2. 6 Operating machines and equipment in energy-efficient modes where applicable 2. 7 Handling materials carefully to reduce wastage and improve accuracy 2. 8 Controlling and reporting leakage, spillage and resource wastage promptly 2. 9 Identifying and segregating different types of waste generated in yarn manufacturing 2. 10 Separating reusable and recyclable materials from non-recyclable waste 2. 11 Handling hazardous waste according to safety guidelines and regulations
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4. 1 Workplace (simulated or actual) 4. 2 Standard operating procedures 4. 3 Workplace documents, signs and symbols 4. 4 Codes of conduct and environmental policy 4. 5 Waste segregation bins and storage area 4. 6 Tools, equipment and facilities appropriate to the process or activities 4. 7 Materials relevant to the proposed activity

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Interpreted green concepts. 1. 2 Minimized resource use in the workplace. 1. 3 Implemented waste management practices.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: INTERPRET PRODUCTIVITY AND QUALITY CONCEPTS IN YARN MANUFACTURING	Nominal Duration: 35 Hrs.	Unit Code: SICIP-TEX-PIQCY-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to interpret productivity and quality concepts in yarn manufacturing. It specifically includes the tasks of interpreting productivity concepts and measures, interpreting quality concepts and standards, identifying factors affecting productivity and quality, and applying basic technical calculations used in a spinning mill.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret productivity concepts and measures	1.1 The concept of <u>productivity</u> and its importance in yarn manufacturing are explained. 1.2 <u>Productivity measures</u> used in a spinning mill are identified and interpreted. 1.3 The difference between production, productivity, efficiency and utilization is explained with examples. 1.4 The relationship between productivity, manufacturing cost and competitiveness is explained.
2. Interpret quality concepts and standards	2.1 The concepts of quality, quality control and quality assurance are explained and differentiated. 2.2 <u>Yarn quality characteristics</u> and their effect on downstream processes are identified. 2.3 Quality standards and reference systems applicable to yarn are interpreted.
3. Identify factors affecting productivity and quality	3.1 <u>Factors affecting productivity and quality</u> at each process stage are identified. 3.2 The influence of raw material, machine, method, manpower and environment is explained. 3.3 Process losses, waste, downtime and rework are identified as productivity losses.
4. Apply basic technical calculations used in a spinning mill	4.1 Yarn numbering systems are identified and converted correctly. 4.2 Hank of sliver and roving is calculated from measured weight and length. 4.3 <u>Basic spinning calculations</u> are performed correctly.

Range of Variables:

Variable	Range (May include but not limited to)
1. Productivity	1.1 Labour productivity (kg of yarn per operator per shift) 1.2 Machine productivity (kg per machine per hour, grams per spindle per shift) 1.3 Material productivity (yarn realization from 100 kg of mixing) 1.4 Energy productivity (kg of yarn per kWh consumed) 1.5 Capital and space productivity 1.6 Partial (single factor) and total factor productivity
2. Productivity measures	2.1 Production per unit time (kg/hour, kg/shift, kg/day) 2.2 Grams per spindle per shift (GPSS) 2.3 Machine efficiency and machine utilization percentage 2.4 Spindle utilization and allocated spindle percentage 2.5 Operator work assignment and workload
3. Factors affecting productivity and quality	3.1 Raw materials 3.2 Workforce 3.3 Machine 3.4 Methods 3.5 Environment
4. Yarn quality characteristics	4.1 Count and count variation (CV%) 4.2 Twist per inch/metre and twist variation 4.3 Strength – count strength product (CSP), tenacity (RKM/cN.tex ⁻¹) and elongation 4.4 Evenness (U% and CV%) 4.5 Imperfections – thin places, thick places and neps per kilometre 4.6 Hairiness index and its variation 4.7 Classimat/objectable faults 4.8 Contamination and foreign fibre level
5. Basic spinning calculations	5.1 Mechanical draft, actual draft and draft constant 5.2 Doubling and total draft in the drafting sequence 5.3 Twist per inch (TPI) and twist multiplier (TM) 5.4 Theoretical (calculated) production of each machine 5.5 Machine efficiency and utilization percentage 5.6 Waste percentage and yarn realization percentage

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Concept, definition and importance of productivity in yarn manufacturing. 1.2 Types and measures of productivity used in a spinning mill. 1.3 Difference between production, productivity, efficiency and utilization.
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	1.4 Concept of quality, quality control, quality assurance and total quality. 1.5 Factors affecting productivity and quality at each stage of the spinning process. 1.6 Sequence of the yarn manufacturing process and the function of each machine. 1.7 Yarn numbering systems, conversions and hank calculation. 1.8 Basic spinning calculations – draft, doubling, twist, production and efficiency. 1.9 Technical terms, symbols and abbreviations used in spinning mills.
2. Underpinning Skills	2.1 Interpreting productivity concepts and measures used in a spinning mill 2.2 Calculating labour, machine, material and energy productivity 2.3 Differentiating production, productivity, efficiency and utilization 2.4 Identifying productivity and quality loss factors at each stage 2.5 Reading and interpreting SOPs, setting sheets, mixing plans and lot particulars 2.6 Performing draft, twist, production and efficiency calculations 2.7 Using a scientific calculator and basic spreadsheet for computation 2.8 Recording and presenting calculated results in prescribed formats
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Classroom and workplace (simulated or actual) 4.2 Computer/laptop with spreadsheet software 4.3 Multimedia projector, screen and audio-visual clips 4.4 Scientific calculator 4.5 Sample fibre, sliver, roving and yarn of different counts 4.6 Wrap reel, wrap block and electronic balance 4.7 SOPs, setting sheets, mixing plans and lot particulars 4.8 Learning manual and reference books

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Interpreted productivity concepts and measures. 1.2 Interpreted quality concepts and standards. 1.3 Identified factors affecting productivity and quality. 1.4 Applied basic technical calculations used in a spinning mill.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM FIBRE AND RAW MATERIAL QUALITY ASSESSMENT	Nominal Duration: 40 Hrs.	Unit Code: SICIP-TEX-PIQCY-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform fibre and raw material quality assessment. It specifically includes the tasks of preparing for fibre and raw material testing, collecting and conditioning samples, testing fibre properties, interpreting fibre test results, and recording and reporting fibre quality data.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for fibre and raw material testing	1.1 The testing requirement is determined from the <u>lot and mixing plan</u> and work instruction. 1.2 <u>Fibre testing instruments</u> are selected and checked for readiness and calibration status. 1.3 Standard atmospheric condition of the laboratory is verified and maintained. 1.4 Test formats, reference standards and specifications are arranged. 1.5 PPE and laboratory safety requirements are applied.
2. Collect and condition samples	2.1 The <u>sampling procedure</u> for bales and mixing is followed correctly. 2.2 Representative samples are drawn from the specified locations and depths. 2.3 Samples are identified, labelled and made traceable to the bale and lot. 2.4 Samples are conditioned for the required period before testing. 2.5 Samples are handled and preserved to avoid contamination, damage and moisture change.
3. Test fibre properties	3.1 <u>Fibre properties</u> are tested following the applicable <u>test methods</u> . 3.2 Instruments are operated as per the operating procedure and manufacturer's manual. 3.3 The required number of readings is taken and abnormal readings are repeated. 3.4 Trash and contamination content is determined. 3.5 Moisture content and regain of the raw material are determined.
4. Interpret fibre test results	4.1 Test results are compared with <u>specification and standard values</u> . 4.2 The effect of fibre properties on yarn quality and spinning performance is explained.

Elements of Competency	Performance Criteria
	4.3 Bale grouping and mixing recommendations are provided to the supervisor.
5. Record and report fibre quality data	5.1 Test data are recorded in the <u>prescribed formats</u> accurately and legibly. 5.2 Bale-wise and lot-wise summaries, averages and CV% are prepared. 5.3 Records are filed and preserved for traceability and audit. 5.4 Instruments, samples and the work area are cleaned and left in order.

Range of Variables:

Variable	Range (May include but not limited to)
1. Lot and mixing plan	1.1 Bale laydown plan and bale grouping 1.2 Mixing percentage of different origins and grades 1.3 Blend ratio (cotton/polyester/viscose and other blends) 1.4 Target count, quality level and end use of the lot
2. Fibre testing instruments	2.1 High Volume Instrument (HVI) system 2.2 Advanced Fibre Information System (AFIS) or equivalent 2.3 Micronaire (fineness) tester 2.4 Fibrograph, comb sorter or Baer sorter for length measurement 2.5 Pressley/Stelometer or bundle strength tester 2.6 Electronic balance (0.001 g accuracy)
3. Sampling procedure	3.1 Percentage and number of bales sampled per lot 3.2 Sampling location and depth within a bale 3.3 Number, size and mass of samples 3.4 Blending and reduction of the gross sample to the test sample 3.5 Sample identification, labelling and traceability 3.6 Conditioning at 20 ± 2 °C and $65 \pm 2\%$ RH 3.7 Minimum conditioning time before testing
4. Fibre properties	4.1 Length – upper half mean length (UHML), mean length, uniformity index, short fibre index 4.2 Fineness – micronaire value, denier/decitex for man-made fibre 4.3 Strength (g/tex) and elongation percentage 4.4 Colour (Rd, +b), grade and appearance 4.5 Trash and leaf content 4.6 Neps per gram (seed coat and fibre neps) 4.7 Moisture content and regain 4.8 Staple length, crimp and lustre of man-made fibre 4.9 Spinning Consistency Index (SCI)

Variable	Range (May include but not limited to)
5. Test methods	5.1 HVI and AFIS operating procedures 5.2 Manual length measurement (comb sorter/Baer sorter) method 5.3 Manual nep counting and appearance assessment
6. Specification and standard values	6.1 HVI/USTER reference values for the origin and grade 6.2 Fibre quality index and spinning consistency index norms
7. Prescribed formats	7.1 Bale-wise test record 7.2 Lot summary and mixing quality sheet 7.3 Bale laydown and mixing record 7.4 Trash, moisture and contamination report 7.5 Laboratory register and instrument log book

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Types, sources and classification of textile fibres used in spinning. 1.2 Physical and chemical properties of cotton and man-made fibres. 1.3 Importance of raw material quality in yarn productivity, quality and cost. 1.4 Bale management, laydown and mixing principles. 1.5 Principles of sampling and representative sample collection. 1.6 Standard atmospheric conditions and the need for conditioning. 1.7 Fibre testing instruments – working principle, operation, care and calibration. 1.8 Standard test methods for length, fineness, strength, maturity, trash and moisture. 1.9 Relationship between fibre properties and yarn strength, evenness, imperfections and hairiness. 1.10 Effect of raw material on end breakage rate, waste percentage and yarn realization.
2. Underpinning Skills	2.1 Preparing the laboratory and instruments for fibre testing 2.2 Drawing representative samples from bales and mixing 2.3 Labelling, conditioning and preserving samples 2.4 Operating HVI/AFIS and manual fibre testing instruments 2.5 Determining fibre length, fineness, strength, maturity and colour 2.6 Determining trash content, nep content and moisture regain 2.7 Interpreting fibre test reports for bale grouping and mixing decisions 2.8 Recording, summarizing and reporting fibre quality data 2.9 Maintaining cleanliness, care and safe handling of testing instruments

3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Testing laboratory with standard atmospheric control 4.2 HVI/AFIS or equivalent fibre testing system 4.3 Shirley trash analyser 4.4 Moisture meter and hot air oven 4.5 Electronic balance (0.001 g accuracy) 4.6 Fibre samples of different origins, grades and types 4.7 Bale sampling tools, sample bags and labels 4.8 PPE and laboratory safety equipment 4.9 Learning manual and audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Prepared for fibre and raw material testing. 1.2 Collected and conditioned representative samples. 1.3 Tested fibre properties following the applicable test methods. 1.4 Interpreted fibre test results against specification and standard values. 1.5 Recorded and reported fibre quality data.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM IN-PROCESS QUALITY CONTROL IN SPINNING PREPARATORY PROCESSES	Nominal Duration: 45 Hrs.	Unit Code: SICIP-TEX-PIQCY-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform in-process quality control in spinning preparatory processes. It specifically includes the tasks of preparing for in-process quality control, checking blow room, lap and chute feed quality, checking card sliver quality, checking draw frame, comber and roving quality, and recording and reporting in-process quality data.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for in-process quality control	1.1 The control plan and testing schedule are interpreted correctly. 1.2 <u>In-process testing instruments</u> and accessories are arranged and checked for readiness. 1.3 Process stages, machines and sampling points are identified. 1.4 Standard values and <u>tolerance limits</u> for each stage are obtained from workplace documents. 1.5 Safety procedures for working on and near running machines are followed.
2. Check blow room, lap and chute feed quality	2.1 Samples are collected from the blow room delivery as per the schedule. 2.2 Lap weight, chute feed weight and their variation are checked against the standard. 2.3 <u>Trash and cleaning efficiency</u> of the blow room line is determined. 2.4 Nep increase and indications of fibre damage are checked.
3. Check card sliver quality	3.1 Card sliver hank and weight per unit length are checked at the specified frequency. 3.2 Sliver evenness (A%, U%, CV%) is checked using the applicable instrument. 3.3 <u>Card quality parameters</u> are checked against the standard. 3.4 Cleaning efficiency and waste percentage of the card are determined. 3.5 Card-to-card and delivery-to-delivery variation is compared. 3.6 Out-of-limit results are reported and corrective action is followed up.

Elements of Competency	Performance Criteria
4. Check draw frame, comber and roving quality	4.1 Sliver hank and weight variation at breaker and finisher drawing are checked. 4.2 <u>Sliver evenness and spectrogram</u> results are examined for periodic and random faults. 4.3 Auto-leveller performance is verified against the standard. 4.4 Comber lap quality, noil percentage and combed sliver quality are checked. 4.5 Roving hank, twist multiplier and unevenness are checked against the standard. 4.6 Roving bobbin build, tension and breakage rate are observed and recorded.
5. Record and report in-process quality data	5.1 All test results are recorded in the <u>prescribed formats</u>. 5.2 Machine-wise and shift-wise summaries are prepared correctly. 5.3 Reports are submitted to the authority within the required time. 5.4 Non-conformances are escalated through the correct reporting channel. 5.5 Records are maintained legibly, completely and traceably.

Range of Variables:

Variable	Range (May include but not limited to)
1. In-process testing instruments	1.1 Wrap reel and wrap block for sliver, roving and yarn 1.2 Digital lap and sliver weighing scale 1.3 Electronic balance (0.01 g and 0.001 g accuracy) 1.4 Evenness tester (capacitive or optical) with spectrogram facility 1.5 Trash analyser 1.6 Nep counting board and hand lens 1.7 Twist tester 1.8 Stop watch and tachometer
2. Tolerance limits	2.1 Hank/count variation limits at each stage 2.2 A%, U% and CV% limits for lap, sliver and roving 2.3 CVb% between deliveries and between machines 2.4 Trash percentage and cleaning efficiency limits 2.5 Nep limit per gram at card and comber 2.6 Waste and noil percentage range 2.7 Twist multiplier range for roving 2.8 Breakage rate limit at each stage
3. Trash and cleaning efficiency	3.1 Trash percentage in mixing and in blow room delivery 3.2 Cleaning efficiency percentage of each machine and of the line 3.3 Waste percentage of each machine

Variable	Range (May include but not limited to)
	3.4 Trash particle size distribution 3.5 Nep generation across the line
4. Card quality parameters	4.1 Sliver hank and hank variation 4.2 Neps per gram in card sliver 4.3 Trash percentage in card sliver 4.4 Flat strip, lick-in and cylinder waste percentage 4.5 Cleaning efficiency of the card 4.6 Short fibre content in card sliver 4.7 Sliver A%, U% and CV% 4.8 Card clothing condition and wire sharpness
5. Sliver evenness and spectrogram	5.1 U% and CV% of sliver and roving 5.2 Spectrogram – mechanical waves and drafting waves 5.3 A% (deviation from nominal hank) 5.4 Thick and thin places in sliver 5.5 Count/hank CV over 1 m, 3 m, 10 m and 50 m
6. Prescribed formats	6.1 Hank/count check sheet 6.2 Evenness and spectrogram test report 6.3 Trash, waste and cleaning efficiency report 6.4 Noil percentage and comber performance record 6.5 Roving quality and breakage record 6.6 Control chart sheet 6.7 Shift quality summary and handover log

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Objectives and principles of in-process quality control in a spinning mill. 1.2 Sequence of preparatory processes and the quality function of each machine. 1.3 Sampling techniques and sampling points for lap, sliver and roving. 1.4 Working principle, operation and care of in-process testing instruments. 1.5 Methods of determining hank, weight variation, evenness and CV%. 1.6 Trash analysis, cleaning efficiency and lint loss calculation. 1.7 Nep formation, nep counting and factors affecting nep level. 1.8 Comber performance parameters – noil percentage, lap quality and combed sliver quality. 1.9 Roving quality parameters – hank, twist multiplier, unevenness and bobbin build. 1.10 Common process faults, their probable causes and corrective actions.
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2. Underpinning Skills	2.1 Interpreting the control plan and testing schedule 2.2 Selecting, checking and using in-process testing instruments 2.3 Collecting representative samples of lap, sliver and roving 2.4 Measuring hank, weight per unit length and weight variation 2.5 Operating the evenness tester and obtaining U%, CV%, A% and spectrogram 2.6 Determining trash percentage, cleaning efficiency and lint loss 2.7 Counting neps and assessing sliver appearance 2.8 Determining comber noil percentage and lap quality 2.9 Checking roving hank, twist and bobbin build 2.10 Recording, summarizing and reporting in-process quality data
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) with preparatory machinery 4.2 Blow room line, carding machine, draw frame, lap former and comber 4.3 Speed frame (simplex) 4.4 Testing laboratory with standard atmospheric control 4.5 Wrap reel, wrap block and digital sliver/lap weighing scale 4.6 Electronic balance (0.001 g and 0.01 g accuracy) 4.7 Evenness tester with spectrogram facility 4.8 Twist tester, tachometer and stop watch 4.9 Hygrometer, thermometer and lux meter 4.10 Sample lap, sliver and roving of different counts 4.11 Control plan, SOPs, standards and recording formats 4.12 PPE and learning manual with audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Prepared for in-process quality control. 1.2 Checked blow room, lap and chute feed quality. 1.3 Checked card sliver quality.
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	1.4 Checked draw frame, comber and roving quality. 1.5 Recorded and reported in-process quality data.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM YARN QUALITY TESTING AND EVALUATION	Nominal Duration: 45 Hrs.	Unit Code: SICIP-TEX-PIQCY-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform yarn quality testing and evaluation. It specifically includes the tasks of preparing the laboratory and instruments for yarn testing, testing yarn count, twist and moisture regain, testing yarn strength and elongation, testing yarn evenness, imperfections and hairiness, and interpreting, recording and reporting yarn test results.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare the laboratory and instruments for yarn testing	1.1 <u>Yarn testing instruments</u> are selected, cleaned and checked for readiness. 1.2 Instrument calibration status and validity are verified before testing. 1.3 Standard atmospheric condition (20 ± 2 °C and $65 \pm 2\%$ RH) is verified and maintained. 1.4 Samples are drawn as per the <u>sampling procedure</u> and conditioned for the required period. 1.5 Laboratory safety and housekeeping requirements are followed.
2. Test yarn count, twist and moisture regain	2.1 Yarn count is determined using the wrap reel and electronic balance. 2.2 Count CV% and deviation from nominal count are calculated. 2.3 Twist per inch/metre is determined following the twist testing method. 2.4 Twist multiplier and twist CV% are calculated. 2.5 Moisture regain is determined and count is corrected to standard regain.
3. Test yarn strength and elongation	3.1 Lea strength and count strength product (CSP) are determined. 3.2 Single yarn strength, tenacity (RKM/cN.tex^{-1}) and elongation are determined. 3.3 Mean, CV%, minimum value and breaking elongation are calculated. 3.4 Results are compared with the standard and the customer requirement.
4. Test yarn evenness, imperfections and hairiness	4.1 Yarn evenness (U% and CV%) is measured using the evenness tester. 4.2 <u>Imperfections</u> are measured at the specified sensitivity levels.

Elements of Competency	Performance Criteria
	4.3 Hairiness index (H) and its standard deviation are determined. 4.4 <u>Diagram, spectrogram and variance-length curve</u> are obtained and examined. 4.5 Test conditions (speed, test length and number of tests) are set as per the standard.
5. Interpret, record and report yarn test results	6.1 Test results are compared with <u>reference standards</u> and the customer specification. 6.2 Deviations and non-conformities are identified and reported promptly. 6.3 Results are recorded in the <u>prescribed formats</u> and plotted on control charts.

Range of Variables:

Variable	Range (May include but not limited to)
1. Yarn testing instruments	1.1 Wrap reel and wrap block 1.2 Electronic balance (0.001 g accuracy) 1.3 Twist tester (direct counting and untwist-retwist type) 1.4 Lea strength tester (CSP) 1.5 Single yarn strength tester (CRE type) with elongation measurement 1.6 Evenness tester with imperfection and spectrogram facility 1.7 Hairiness tester or hairiness module 1.8 Yarn fault classification system (Classimat or equivalent) 1.9 Black board winder and appearance grading board/cabinet 1.10 Moisture meter and hot air oven 1.11 Package and cone weighing scale
2. Sampling procedure	2.1 Number of packages/cones sampled per lot and per machine 2.2 Number of tests per package and total number of tests 2.3 Sample identification, labelling and traceability 2.4 Conditioning time and standard testing atmosphere
3. Imperfections	3.1 Thin places at -30%, -40% and -50% sensitivity 3.2 Thick places at +35% and +50% sensitivity 3.3 Neps at +140%, +200% and +280% sensitivity 3.4 Total imperfection index (IPI) per kilometre
4. Diagram, spectrogram and variance-length curve	4.1 Mass variation diagram at different cut lengths 4.2 Spectrogram showing mechanical and drafting waves 4.3 CV% at 1 m, 3 m, 10 m, 50 m and 100 m cut lengths

Variable	Range (May include but not limited to)
5. Reference standards	6.1 USTER STATISTICS percentile values (5%, 25%, 50%, 75%, 95%) 6.2 ASTM, ISO and BS standard test methods 6.3 Buyer specification, contract and end-use requirement 6.4 Mill in-house norms and control limits 6.5 Previous lot results and trend data
6. Prescribed formats	7.1 Count, twist and strength test report 7.2 Evenness, imperfection and hairiness test report 7.3 Classimat/fault classification report 7.4 Lot-wise quality summary and quality certificate 7.5 Control chart sheet and trend char 7.6 Laboratory register and instrument log book

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Objectives and importance of yarn quality testing and evaluation. 1.2 Standard atmospheric conditions, conditioning and their effect on test results. 1.3 Sampling principles and sample size determination for yarn testing. 1.4 Working principle, operation, care and calibration of yarn testing instruments. 1.5 Standard test methods for count, twist, strength, elongation, evenness and hairiness. 1.6 Concept of evenness, index of irregularity, imperfections and hairiness. 1.7 USTER STATISTICS and the use of percentile values for benchmarking. 1.8 Effect of yarn quality characteristics on weaving, knitting and dyeing performance. 1.9 Preparation of test reports, quality certificates and documentation requirements.
2. Underpinning Skills	2.1 Preparing the laboratory and verifying atmospheric conditions 2.2 Verifying instrument calibration status before testing 2.3 Drawing, labelling and conditioning yarn samples 2.4 Determining yarn count using wrap reel and electronic balance 2.5 Determining twist per inch and twist multiplier 2.6 Determining lea strength, CSP, single yarn strength, tenacity and elongation 2.7 Operating the evenness tester and obtaining U%, CV% and imperfections

	2.8 Comparing results with USTER STATISTICS and customer specification 2.9 Calculating mean, standard deviation, CV% and confidence limits 2.10 Preparing test reports
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Yarn testing laboratory with standard atmospheric control 4.2 Wrap reel, wrap block and electronic balance (0.001 g accuracy) 4.3 Twist tester 4.4 Lea strength tester and single yarn strength tester (CRE) 4.5 Evenness tester with imperfection, spectrogram and hairiness facility 4.6 Black board winder and appearance grading cabinet with standard photographs 4.7 Moisture meter 4.8 Conditioning chamber 4.9 Yarn samples of different counts, blends and quality levels 4.10 Standard test methods, specification sheets and recording formats 4.11 Computer with spreadsheet and reporting facility 4.12 PPE and learning manual with audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Prepared the laboratory and instruments for yarn testing. 1.2 Tested yarn count, twist and moisture regain. 1.3 Tested yarn strength and elongation. 1.4 Tested yarn evenness, imperfections and hairiness. 1.5 Interpreted, recorded and reported yarn test results.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)

3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.
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Unit of Competency: IDENTIFY AND CONTROL YARN FAULTS AND CONTAMINATION	Nominal Duration: 35 Hrs.	Unit Code: SICIP-TEX-PIQCY-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to identify and control yarn faults and contamination. It specifically includes the tasks of identifying yarn, package and process faults, analysing causes of faults, controlling contamination and foreign fibre, and applying corrective and preventive actions.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify yarn, package and process faults	1.1 <u>Yarn faults</u> are identified visually and from test data. 1.2 <u>Package faults</u> are identified during winding, conditioning and packing. 1.3 Fault frequency, severity and distribution are assessed and classified. 1.4 Faulty material is segregated, marked and held as per workplace procedure.
2. Analyse causes of faults	2.1 Fault samples are examined and traced to the <u>originating process or machine</u> . 2.2 Machine, material, method, manpower and environment causes are separated. 2.3 Spindle-wise, machine-wise and shift-wise data are compared to locate the source.
3. Control contamination and foreign fibre	3.1 <u>Sources of contamination</u> are identified at each stage of the process. 3.2 <u>Contamination control measures</u> are implemented in the work area. 3.3 Contamination sorting and removal practices are monitored and verified. 3.4 Performance of the foreign fibre detector and yarn clearer is monitored.
4. Apply corrective and preventive actions	4.1 <u>Corrective actions</u> are recommended and implemented within the level of authority. 4.2 Preventive measures are proposed to avoid recurrence of the fault. 4.3 Machine settings, accessories and schedules are corrected through the authorized channel. 4.4 Affected material is reprocessed, downgraded or rejected as per procedure. 4.5 Operators are briefed on the correct work method where the cause is human.

Range of Variables:

Variable	Range (May include but not limited to)
1. Yarn faults	1.1 Thick places, thin places and neps 1.2 Slubs and spun-in fly 1.3 Count variation and mixed count 1.4 High and uneven hairiness 1.5 Foreign fibre and polypropylene contamination 1.6 Oil, grease and dirt stains 1.7 Double ends and multiple ends 1.8 Hard twist, soft twist and untwisted places 1.9 Barré-causing variation between machines and lots 1.10 Improper splice and bad knot
2. Package faults	2.1 Soft package and hard package 2.2 Bulged, dished, conical and saddle-shaped package 2.3 Damaged nose, base and cone body 2.4 Wrong package weight and length 2.5 Mixed count or mixed lot in a carton 2.6 Loose, missing or incorrect labelling 2.7 Improper conditioning of the package
3. Originating process or machine	3.1 Mixing and blow room 3.2 Carding 3.3 Breaker and finisher drawing 3.4 Lap former and comber 3.5 Speed frame (simplex) 3.6 Ring frame 3.7 Winding and yarn clearing 3.8 Conditioning, packing and dispatch 3.9 Humidification and utility plant 3.10 Material handling, storage and transport
4. Sources of contamination	4.1 Raw cotton, bale wrappers and bale straps 4.2 Packing materials (jute, polypropylene, plastic, paper) 4.3 Operator clothing, hair, buttons and personal items 4.4 Cleaning cloth, mops, brooms and dusters 4.5 Residue of the previous lot in machines, cans and trolleys 4.6 Oil, grease, rust and paint flakes 4.7 Insects, dust, fly and airborne fibre 4.8 Storage, transport and handling areas 4.9 Waste re-mixing and floor sweepings
5. Contamination control measures	5.1 Manual contamination sorting at the pre-opening/mixing stage 5.2 Use of cotton-only cleaning cloth and lint-free wipers 5.3 Controlled work clothing, hair covering and prohibition of loose items 5.4 Colour-coded and dedicated cans, trolleys and containers 5.5 Machine and can cleaning between lots and counts

Variable	Range (May include but not limited to)
	5.6 Contamination/foreign fibre detector at blow room and winding 5.7 Operator awareness, training and reward system 5.8 Regular contamination audit and monitoring 5.9 Segregation and controlled reuse of waste
6. Corrective actions	6.1 Correcting machine settings, gauges and drafting parameters 6.2 Replacing worn cots, aprons, travellers, rings, spindle tapes and card clothing 6.3 Adjusting yarn clearer settings and splicer parameters 6.4 Adjusting draft, twist and speed within permissible limits 6.5 Correcting temperature, relative humidity and air movement 6.6 Retraining operators on piecing, doffing and handling methods 6.7 Revising cleaning, lubrication and maintenance schedules 6.8 Segregating, reprocessing or downgrading defective material 6.9 Improving material handling, storage and identification practices

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Classification of yarn faults – frequent, seldom-occurring and objectionable. 1.2 Appearance, causes and effects of each type of yarn and package fault. 1.3 Fault tracing methods and identification of the originating machine. 1.4 Contamination control systems, detectors and clearing devices. 1.5 Yarn clearer types, clearing curve and setting principles. 1.6 Corrective and preventive action procedures and documentation. 1.7 Reporting, escalation and standardization requirements.
2. Underpinning Skills	2.1 Identifying yarn faults visually and from test data 2.2 Identifying package faults during winding and packing 2.3 Segregating, marking and preserving faulty material and samples 2.4 Tracing faults to the originating process, machine or spindle 2.5 Implementing and monitoring contamination control measures 2.6 Monitoring foreign fibre detector and yarn clearer performance

	2.7 Measuring and reporting the contamination level 2.8 Recommending and implementing corrective and preventive actions 2.9 Preparing fault analysis
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) with ring frame and winding machine 4.2 Yarn fault sample board and reference fault samples 4.3 Contamination and foreign fibre samples 4.4 Black board winder and appearance grading cabinet 4.5 Evenness tester with spectrogram facility 4.6 Yarn fault classification system (Classimat or equivalent) 4.7 Yarn clearer and splicer with setting facility 4.8 Hand lens, magnifier and inspection light 4.9 Colour-coded containers, trolleys and contamination sorting table 4.10 Computer with spreadsheet and charting facility 4.11 PPE and learning manual with audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Identified yarn, package and process faults. 1.2 Analysed causes of faults using appropriate techniques. 1.3 Controlled contamination and foreign fibre in the work area. 1.4 Applied corrective and preventive actions.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: MEASURE AND IMPROVE MACHINE AND MANPOWER PRODUCTIVITY	Nominal Duration: 45 Hrs.	Unit Code: SICIP-TEX-PIQCY-06-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to measure and improve machine and manpower productivity. It specifically includes the tasks of collecting production and performance data, calculating production, efficiency and utilization, analysing downtime, breakages and losses, and monitoring manpower productivity and work assignment.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Collect production and performance data	1.1 <u>Production data</u> are collected from machines, records and monitoring systems. 1.2 Data collection frequency, format and responsibility are followed as per procedure. 1.3 Data collection methods and instruments are used correctly.
2. Calculate production, efficiency and utilization	2.1 Theoretical (calculated) production of each machine is computed using the applicable <u>production formulae</u> . 2.2 Actual production is obtained and compared with the theoretical production. 2.3 <u>Efficiency and utilization measures</u> are calculated correctly. 2.4 Grams per spindle per shift and kilograms per machine per hour are calculated.
3. Analyse downtime, breakages and losses	3.1 <u>Downtime and stoppage causes</u> are recorded and classified using reason codes. 3.2 End breakage rate is measured by the standard method and expressed per 100 spindle hours. 3.3 <u>Causes of end breakage</u> are analysed and ranked by frequency. 3.4 Idle spindles, idle machines and lost production are quantified.
4. Monitor manpower productivity and work assignment	4.1 Standard time and workload for each job are determined or verified. 4.2 Operator work assignment (machine/spindle allocation) is checked against the norm. 4.3 Material handling, movement and waiting time are analysed.

Range of Variables:

Variable	Range (May include but not limited to)
1. Production data	1.1 Machine speed (rpm, m/min) and delivery rate 1.2 Count/hank being produced and twist multiplier 1.3 Number of spindles, deliveries or heads in operation 1.4 Running time, stoppage time and available time 1.5 Doff weight, doff time and number of doffs per shift 1.6 Waste quantity collected at each machine 1.7 End breakage count and breakage survey data 1.8 On-line production monitoring/mill information system data
2. Production formulae	2.1 Theoretical production of card, draw frame, comber, speed frame, ring frame and winder 2.2 Production in kg per hour, per shift and per day 2.3 Grams per spindle per shift (GPSS) and kg per spindle per shift 2.4 Conversion of hank/count and length to weight 2.5 Machine efficiency percentage and utilization percentage 2.6 Production loss due to stoppages and idle spindles
3. Efficiency and utilization measures	3.1 Machine efficiency (actual production ÷ calculated production) 3.2 Machine utilization (running time ÷ available time) 3.3 Spindle utilization and allocated spindle percentage 3.4 Doffing efficiency and creeling efficiency 3.5 Yarn realization percentage
4. Downtime and stoppage causes	4.1 Power failure and voltage fluctuation 4.2 Mechanical breakdown and electrical breakdown 4.3 Doffing, creeling and lot change 4.4 Count change and setting change 4.5 Waiting for maintenance or spares 4.6 Scheduled cleaning and preventive maintenance 4.7 Absenteeism and manpower shortage 4.8 Quality-related stoppage and rework
5. Causes of end breakage	5.1 Poor raw material, high short fibre and high nep level 5.2 Incorrect draft distribution, twist and spindle speed 5.3 Defective or worn cots, aprons, travellers, rings and spindles 5.4 Incorrect traveller number and clearer setting 5.5 Roving bobbin defects and creel problems 5.6 Improper spacer, condenser and top arm pressure 5.7 Unsuitable temperature and relative humidity 5.8 Excessive fly, dust and lint accumulation 5.9 Poor piecing quality and operator handling 5.10 Vibration, eccentricity and misalignment

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Concept and importance of productivity measurement in a spinning mill. 1.2 Sources and types of production and performance data. 1.3 Methods and instruments used for production data collection. 1.4 Calculation of theoretical production for each machine in the spinning line. 1.5 Machine efficiency, utilization, GPSS and OEE calculation methods. 1.6 Classification and analysis of downtime and machine stoppages. 1.7 End breakage survey methods and calculation per 100 spindle hours. 1.8 Causes of end breakage and their effect on productivity and quality. 1.9 Operator work assignment, workload calculation and machine allocation norms. 1.10 Effect of productivity measures on yarn quality and the need for verification.
2. Underpinning Skills	2.1 Collecting production and performance data from machines and records 2.2 Using tachometer, stop watch, counter and weighing devices correctly 2.3 Calculating theoretical and actual production of each machine 2.4 Calculating machine efficiency, utilization, GPSS and OEE 2.5 Recording and classifying downtime using reason codes 2.6 Conducting an end breakage survey and calculating the breakage rate 2.7 Analysing causes of end breakage and ranking them by frequency 2.8 Quantifying idle spindles, idle machines and lost production 2.9 Calculating standard time, workload and operator allocation 2.10 Measuring, comparing and reporting improvement results
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) with spinning machinery in operation

	4.2 Tachometer, stopwatch and counter 4.3 Electronic balance and platform weighing scale 4.4 Production monitoring system or machine display data 4.5 Shift log books, downtime records and production formats 4.6 End breakage survey formats and clipboards 4.7 Work study formats – time study sheet, work sampling sheet, process chart 4.8 Computer with spreadsheet and charting facility 4.9 Calculator and measuring tape 4.10 PPE and learning manual with audio-visual clips
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Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Collected production and performance data. 1.2 Calculated production, efficiency and utilization. 1.3 Analysed downtime, end breakages and production losses. 1.4 Monitored manpower productivity and work assignment.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY WASTE CONTROL AND COST REDUCTION TECHNIQUES	Nominal Duration: 30 Hrs.	Unit Code: SICIP-TEX-PIQCY-07-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply waste control and cost reduction techniques in yarn manufacturing. It specifically includes the tasks of identifying and classifying waste, measuring and calculating waste and yarn realization, analysing causes of excess waste, applying waste control and reuse measures, and reporting waste and cost saving.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify and classify waste generated in the spinning process	1.1 <u>Types of waste</u> generated at each process stage are identified correctly. 1.2 Reusable, saleable and non-usable waste are classified and separated. 1.3 Waste collection points and containers are identified and used correctly.
2. Measure and calculate waste and yarn realization	2.1 Waste is collected, weighed and recorded stage-wise at the specified frequency. 2.2 <u>Waste percentage</u> of each machine and department is calculated correctly. 2.3 <u>Yarn realization</u> percentage is calculated for the lot. 2.4 Invisible loss is determined and verified against the material balance. 2.5 Results are compared with the <u>waste norms</u> and deviations are identified.
3. Analyse causes of excess waste	3.1 Machines and processes exceeding the norm are identified from the data. 3.2 Causes of excess waste are analysed using appropriate techniques. 3.3 Machine settings, accessory condition and work practices are examined. 3.4 Findings are discussed and verified with the concerned personnel. 3.5 Priority areas for waste reduction are identified and ranked.
4. Apply waste control and reuse measures	4.1 <u>Waste control measures</u> are implemented within the level of authority. 4.2 Machine settings and waste extraction are corrected through the authorized channel.

Elements of Competency	Performance Criteria
	4.3 Soft waste reuse in mixing is carried out within the permissible limit. 4.4 The effect of waste control measures on yarn quality is verified. 4.5 Waste handling, storage and disposal are done as per the environmental procedure.
5. Report waste and cost saving	5.1 Cost elements of yarn manufacturing are identified. 5.2 The cost impact of waste, downtime and rejection is estimated in simple terms. 5.3 Savings achieved from improvement measures are calculated. 5.4 Waste and cost reports are prepared in the prescribed format and submitted. 5.5 Records are maintained for traceability, audit and trend analysis.

Range of Variables:

Variable	Range (May include but not limited to)
1. Types of waste	1.1 Blow room droppings and filter waste 1.2 Card flat strips, licker-in and cylinder droppings 1.3 Comber noil 1.4 Sliver, lap and roving waste 1.5 Pneumafil and bonda waste at the ring frame 1.6 Ring frame sweep waste and hard waste 1.7 Winding hard waste 1.8 Sweeping and floor waste 1.9 Filter, duct and suction system waste 1.10 Damaged packages, cones, bobbins and cans 1.11 Packing material waste (poly bags, cartons, straps) 1.12 Used oil, grease and oil-contaminated cotton (hazardous)
2. Waste percentage	2.1 Machine-wise and stage-wise waste percentage 2.2 Department-wise waste percentage 2.3 Total waste percentage of the mill 2.4 Soft waste and hard waste percentage 2.5 Saleable waste percentage 2.6 Invisible loss percentage 2.7 Usable and unusable waste percentage
3. Yarn realization	3.1 $\text{Yarn realization} = (\text{yarn produced} \div \text{mixing consumed}) \times 100$ 3.2 Count-wise and lot-wise realization 3.3 Carded and combed yarn realization norms 3.4 Effect of trash, noil and waste on realization 3.5 Effect of moisture regain on realization figures

Variable	Range (May include but not limited to)
4. Waste norms	4.1 Mill in-house norms for each machine and department 4.2 Permissible noil percentage for the count and quality 4.3 Permissible lint loss percentage in droppings 4.4 Tolerance for invisible loss
5. Waste control measures	5.1 Optimization of blow room and card settings and speeds 5.2 Control of noil percentage at the comber 5.3 Reduction of end breakage at speed frame and ring frame 5.4 Control of pneumafil, bonda and hard waste 5.5 Proper handling, storage and identification of material 5.6 Reuse of soft waste within the permissible limit 5.7 Regular checking of suction, ducts and filter systems 5.8 Optimization of yarn clearer settings 5.9 Waste audit, monitoring and trend analysis 5.10 Operator awareness, accountability and training 5.11 Good housekeeping and 5S practices

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Types, sources and classification of waste in a spinning mill. 1.2 Importance of waste control in cost, quality and environmental performance. 1.3 Waste collection, weighing and recording procedures. 1.4 Calculation of machine-wise, department-wise and total waste percentage. 1.5 Concept and calculation of yarn realization and material balance. 1.6 Analysis of good fibre (lint) content in droppings and noil. 1.7 Waste norms and industry benchmarks for carded and combed counts. 1.8 Causes of excess waste at each stage of the process. 1.9 Waste storage, disposal and environmental compliance requirements. 1.10 Waste and cost reporting formats and documentation requirements.
2. Underpinning Skills	2.1 Identifying and classifying waste generated at each stage 2.2 Applying colour-coded segregation and safe handling practices 2.3 Collecting, weighing and recording waste stage-wise 2.4 Calculating machine-wise, department-wise and total waste percentage 2.5 Calculating yarn realization and preparing a material balance 2.6 Analysing lint content in droppings and noil

	2.7 Comparing results with waste norms and identifying deviations 2.8 Analysing causes of excess waste using appropriate techniques 2.9 Verifying the effect of waste control measures on yarn quality 2.10 Estimating the cost impact of waste, downtime and rejection
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) with spinning machinery 4.2 Waste collection bins, colour-coded containers and trolleys 4.3 Platform weighing scale and electronic balance 4.4 Trash analyser for lint and trash analysis 4.5 Material balance, waste report and realization formats 4.6 Waste norms, benchmark data and machine catalogues 4.7 Computer with spreadsheet and charting facility 4.8 Calculator and recording formats 4.9 Environmental procedure, SOPs and safety guidelines 4.10 PPE and learning manual with audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Identified and classified waste generated in the spinning process. 1.2 Measured and calculated waste percentage and yarn realization. 1.3 Analysed causes of excess waste. 1.4 Applied waste control and reuse measures. 1.5 Reported waste and cost saving to the authority.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)

3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.
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Unit of Competency: APPLY QUALITY MANAGEMENT SYSTEM	Nominal Duration: 25 Hrs.	Unit Code: SICIP-TEX-PIQCY-08-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply quality management system. It specifically includes the tasks of interpreting quality management system requirements, participating in quality audit and documentation, and preparing and presenting improvement reports.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret quality management system requirements	1.1 <u>Quality management system (QMS)</u> principles and requirements are interpreted. 1.2 Roles, responsibilities and authorities for quality in the workplace are identified. 1.3 Customer and compliance requirements applicable to the mill are identified.
2. Participate in quality audit and documentation	2.1 <u>Types of audit</u> and the audit process are explained. 2.2 Checklists and records are prepared for the internal audit. 2.3 Corrective and preventive actions are followed up to closure. 2.4 Document and record control requirements are complied with. 2.5 Traceability of material, process and test records is maintained.
3. Prepare and present improvement reports	3.1 Data are compiled, analysed and presented using tables and graphs. 3.2 <u>Improvement reports</u> are prepared in the required format. 3.3 Reports and records are filed and preserved as per the retention requirement.

Range of Variables:

Variable	Range (May include but not limited to)
1. Quality management system (QMS)	1.1 ISO 9001 quality management system requirements 1.2 Quality management principles – customer focus, leadership, engagement of people, process approach, improvement, evidence-based decision making, relationship management 1.3 Total Quality Management (TQM) concept 1.4 Process approach and the plan-do-check-act cycle 1.5 Risk-based thinking (awareness level)

Variable	Range (May include but not limited to)
	1.6 Buyer and customer compliance requirements 1.7 Product/process certification schemes – OEKO-TEX, GOTS, BCI (awareness level)
2. Types of audits	2.1 Internal (first-party) audit 2.2 Second-party (customer/supplier) audit 2.3 Third-party (certification) audit 2.4 Process audit, product audit and system audit 2.5 Housekeeping and 5S audit 2.6 Contamination audit 2.7 Machine and maintenance audit 2.8 Follow-up audit for verification of corrective action
3. Improvement reports	1. Daily, weekly and monthly quality report 2. Productivity and efficiency report 3. Waste and yarn realization report 6. Internal audit report and non-conformance report 7. Management review input report

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Concept, principles and benefits of a quality management system. 1.2 Quality policy, objectives and organizational responsibilities for quality. 1.3 Structure and hierarchy of QMS documentation 1.4 Document control, record control and traceability requirements. 1.5 Sampling plans, inspection levels and acceptance criteria. 1.6 Types of audits, the audit process and audit documentation. 1.7 Data presentation, report writing and presentation techniques.
2. Underpinning Skills	2.1 Interpreting QMS principles, policy and objectives 2.2 Locating and using QMS documentation correctly 2.3 Preparing audit checklists and recording objective evidence 2.4 Preparing and presenting improvement reports to the authority 2.5 Compiling and analysing data using spreadsheets 2.6 Presenting data using tables, graphs and charts 2.7 Preparing and presenting improvement reports to the authority
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties

	3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Classroom and workplace (simulated or actual) 4.2 Computer/laptop with spreadsheet and presentation software 4.3 Multimedia projector, screen and whiteboard 4.4 Scientific calculator 4.5 QMS documentation set – policy, manual, procedures, formats 4.6 Sample process and quality data sets for analysis 4.7 Standards, specifications and reference materials 4.8 Learning manual, case studies and audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Interpreted quality management system requirements. 1.2 Participated in quality audit and documentation. 1.3 Prepared and presented improvement reports.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: MONITOR PROCESS ENVIRONMENT, MACHINE CONDITION AND CALIBRATION	Nominal Duration: 20 Hrs.	Unit Code: SICIP-TEX-PIQCY-09-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to monitor the process environment, machine condition and instrument calibration. It specifically includes the tasks of monitoring atmospheric and workplace conditions, monitoring machine condition and critical settings, verifying calibration and care of testing instruments, and maintaining records.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Monitor atmospheric and workplace conditions	1. 1 <u>Atmospheric parameters</u> are measured at the specified points and frequency. 1. 2 The effect of atmospheric conditions on quality and productivity is explained. 1. 3 Dust, fly, lighting and noise levels are observed and abnormalities are reported.
2. Monitor machine condition and critical settings	2. 1 The condition of <u>spare parts</u> is assessed against the replacement norm. 2. 2 Critical machine settings affecting quality are checked as per the schedule. 2. 3 A machine audit is carried out using the prescribed checklist. 2. 4 Restoration of guards and safety devices after any work is verified.
3. Verify calibration and care of testing instruments	3.1 Calibration requirements and the calibration schedule are interpreted. 3.2 Routine checking with standard or reference material is carried out. 3.3 Calibration certificates and traceability records are preserved.
4. Maintain records	4.1 Monitoring results are recorded in the <u>prescribed formats</u> . 4.2 Follow-up of corrective action, if any deviation, is carried out until closure. 4.3 Records are preserved for traceability and audit.

Range of Variables:

Variable	Range (May include but not limited to)
1. Atmospheric parameters	1. 1 Dry bulb temperature 1. 2 Wet bulb temperature 1. 3 Relative humidity (RH%) 1. 4 Dust and fly concentration 1. 5 Noise level (dB)
2. Spare parts	2. 1 Cots and aprons – hardness, surface condition and buffing schedule 2. 2 Travellers and rings – wear, life and replacement schedule 2. 3 Spindle tapes, bolsters and drives 2. 4 Card clothing – cylinder, doffer, flat and licker-in wire condition 2. 5 Comber needle/unicomb and top comb condition 2. 6 Spacers, condensers and trumpets 2. 7 Lappet hooks, yarn guides and balloon control rings 2. 8 Filters, suction ducts and waste collection systems 2. 9 Belts, pulleys, chains and bearings
3. Prescribed formats	3.1 Humidity and temperature log sheet 3.2 Machine audit checklist 3.3 Machine parts condition and replacement schedule record 3.4 Calibration register and instrument status label 3.5 Deviation and abnormality report 3.6 Maintenance intimation and follow-up record 3.7 Trend chart and monthly monitoring summary

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Importance of the process environment in yarn quality and productivity. 1. 2 Psychrometry and the use of hygrometer, thermometer and psychrometric chart. 1. 3 Recommended atmospheric conditions for each department of a spinning mill. 1. 4 Effect of humidity and temperature on end breakage, fly, waste and yarn quality. 1. 5 Working principle of the humidification plant and its control. 1. 6 Critical machine settings and their effect on quality and productivity. 1. 7 Machine audit procedure and use of audit checklists. 1. 8 Calibration methods, frequency, standards and reference materials. 1. 9 Care, cleaning, handling and storage of testing instruments.
2. Underpinning Skills	2. 1 Measuring temperature, relative humidity and air movement

	2. 2 Using hygrometer, thermometer, lux meter and sound level meter 2. 3 Reporting atmospheric deviations to the utility section 2. 4 Checking critical machine settings using gauges 2. 5 Assessing the condition of cots, aprons, travellers, rings and card clothing 2. 6 Carrying out a machine audit using a checklist 2. 7 Interpreting the calibration schedule and verifying calibration status 2. 8 Cleaning, handling and storing testing instruments correctly 2. 9 Escalating deviations and following up corrective action to closure
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) with spinning machinery and humidification plant 4.2 Whirling/digital hygrometer, thermometer and psychrometric chart 4.3 Anemometer, lux meter and sound level meter 4.4 Dust monitoring device (where available) 4.5 Feeler, setting and thickness gauge set 4.6 Cot hardness (shore) tester and cot buffing machine 4.7 Traveller gauge and traveller sample set 4.8 Standard weights and reference/standard test materials 4.9 Tachometer, stop watch and hand tool set 4.10 Machine audit checklists and calibration register 4.11 Calibration status labels and instrument log books 4.12 PPE and learning manual with audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Monitored atmospheric and workplace conditions. 1. 2 Monitored machine condition and critical settings. 1. 3 Verified calibration and care of testing instruments. 1. 4 Maintained records.
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2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

WORKSHOP / LAB FACILITY STANDARD

Course Name:	Productivity Improvement and Quality Control in Yarn
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/Lab/Classroom cum Workshop):

1. Classroom – 600 sft (56 sqm)
2. Testing laboratory/workshop – 1500 sft (140 sqm), **OR**
3. Classroom cum laboratory/workshop – 2000 sft (186 sqm)

Major Training Equipment and Training Facilities:

S.N.	Major Equipment and Training Facilities	Required Quantity
1.	Computer/Laptop	2
2.	Multimedia projector & screen	1
3.	Printer	1
4.	Scanner	1
5.	Internet connection with Wi-Fi facility	1
6.	Whiteboard with accessories	1
7.	Scientific calculator	5
8.	Standard testing atmosphere control unit (20 ± 2 °C, $65 \pm 2\%$ RH)	1
9.	High Volume Instrument (HVI) or equivalent fibre testing system	1
10.	Advanced Fibre Information System (AFIS) or equivalent	1
11.	Micronaire (fineness) tester	1
12.	Fibrograph/comb sorter/Baer sorter for fibre length	1
13.	Bundle strength tester (Pressley/Stelometer)	1
14.	Trash analyser (Shirley type)	1
15.	Fibre identification kit (burning, solubility, melting point apparatus)	1 set
16.	Stereo/projection microscope with slide preparation set	1
17.	Hot air oven for moisture determination	1
18.	Moisture meter	1
19.	Electronic balance (0.001 g accuracy)	1
20.	Electronic balance (0.01 g accuracy)	2
21.	Platform weighing scale for waste and package weighing	1

S.N.	Major Equipment and Training Facilities	Required Quantity
22.	Wrap reel with wrap block (sliver, roving and yarn)	3
23.	Digital sliver/lap weighing scale	1
24.	Twist tester	2
25.	Lea strength tester (CSP)	1
26.	Single yarn strength tester (CRE type) with elongation measurement	1
27.	Evenness tester with imperfection, spectrogram and hairiness facility	1
28.	Yarn fault classification system (Classimat or equivalent)	1
29.	Black board winder with appearance grading cabinet and standard photographs	1
30.	Yarn conditioning chamber	1
31.	Nep counting board and hand lens/magnifier	5
32.	Whirling/digital hygrometer and thermometer set	5
33.	Anemometer	2
34.	Lux meter	2
35.	Sound level meter	2
36.	Tachometer	3
37.	Stop watch	10
38.	Feeler, setting and thickness gauge set	5
39.	Cot hardness (shore) tester	1
40.	Cot buffing machine	1
41.	Traveller gauge and traveller sample set (assorted)	2 sets
42.	Hand tool set (spanner, screwdriver, plier, nipper, hook)	5
43.	Standard weights and reference/standard test materials	2 sets
44.	Sample set of fibre, lap, sliver, roving and yarn of different counts	As required
45.	Yarn fault and contamination reference sample board	3
46.	Colour coded waste segregation bins and containers	6
47.	Material handling trolley	2
48.	Industrial vacuum cleaner/air blowing gun	2
49.	Tool board, storage cabinet and sample storage rack	3
50.	Fire extinguisher	2

The following conditions must be fulfilled –

1. The institute shall not use the same facilities for any other projects/organizations offering a similar course.
2. The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facilities required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirements of the Competency Standard (CS) provided by SICIP.

For the operation of the training course on Productivity Improvement and Quality Control in Yarn, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

This occupation requires observation, measurement and analysis to be carried out on running production machinery. Where a training institute does not possess a full-scale spinning line and a fully equipped yarn testing laboratory, the institute must establish a formal linkage with a spinning mill so that the practical sessions of the occupation-specific units – particularly in-process quality control, yarn testing, fault and contamination control, productivity measurement and waste control – can be conducted in an actual workplace under supervision.

The institute will also provide all other hand tools, instruments and consumables as per the CS for 25 trainees. The institute will also arrange adequate seating arrangements and classroom setup for the 25 trainees.